

Project Proposal:

When Approximate Reasoning from Data Beats Classical Deduction

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1 Problem Statement

Many rules are not universally true but are typically true under a distribution of examples. Exceptions are common in real life; contextual information may influence a previously plausible conclusion, and exact logical entailment can be too fragile or too computationally expensive to serve as the main reasoning primitive. As a result, we shift from classical deduction to approximate reasoning from data: instead of asking whether a rule is always valid, we ask whether it holds with high probability on examples drawn from a distribution (Valiant, 1984; Valiant, 2000; Khardon and Roth, 1997).

This project proposes an empirical study of that shift. We will compare two views of reasoning:

1. **Classical deduction:** a rule is useful only if it is universally valid.
2. **Approximate reasoning:** a rule is useful if it has high empirical validity on sampled data, even in the presence of exceptions.

Our main goal is to investigate when approximate reasoning from examples remains informative and computationally tractable even when exact deduction is either uninformative or computationally difficult. We will study this question primarily using synthetic Boolean data, where the underlying distribution, exception rates, and contextual refinements are fully controlled. We will then complement the synthetic experiments with one or two real-data case studies, using the UCI Mushroom dataset and, if time permits, the EPA AirData AQI files.

2 Related Work

The theoretical foundation for this project comes from the Probably Approximately Correct (PAC) learning framework introduced by Valiant. PAC learning formalized learning from examples under probabilistic guarantees and shifted attention from perfect identification to approximately correct generalization (Valiant, 1984). This perspective is directly relevant here because our project studies

rules that may fail on some examples yet remain useful when they hold with sufficiently high probability.

Khardon and Roth extended this perspective by proposing a "learning to reason" framework, in which reasoning tasks are supported by data rather than by symbolic knowledge alone (Khardon and Roth, 1997). Their work motivates the basic comparison in our project: exact logical validity may be too restrictive, while empirical validity estimated from examples may support effective reasoning.

Valiant later developed the framework of robust logics, arguing that logical inference should be adapted to noisy and imperfectly learned rules (Valiant, 2000). This is especially relevant to our synthetic experiments, where we will explicitly introduce exceptions and observe when approximate rules remain stable enough to support reasoning.

For the non-monotonic component of our project, we referenced Goldszmidt and Pearl, who studied reasoning with defaults and varying strengths of belief in tractable non-monotonic systems (Goldszmidt and Pearl, 1991). Their work suggests that adding more specific information can change a reasonable conclusion, a behavior we plan to test empirically through conditional rule validity.

For the abductive component, we referenced Juba’s formulation of learning abductive reasoning from random examples, where plausible explanations are evaluated directly from data rather than from a fully specified symbolic theory (Juba, 2016). Our project will use the same high-level idea: explanations should be judged by how well they account for observed outcomes and how frequently they succeed on the data.

Finally, our project differs from prior theory-focused work by emphasizing a hybrid empirical design, involving controlled synthetic Boolean data for conceptual tests and real datasets that let us examine whether these reasoning phenomena remain visible outside hand-crafted examples.

3 Proposed Approach

Overall Experimental Design

The project will have three components:

1. **Synthetic Boolean experiments (main component):** controlled distributions over Boolean variables with planted approximate rules, exceptions, and contextual refinements.
2. **UCI Mushroom case study:** a fully categorical real dataset for evaluating approximate validity and rule exceptions in a structured domain.
3. **EPA AQI case study:** a real environmental dataset used to examine contextual validity across time, seasonality, and geographic subgroups.

Synthetic Boolean World

We will generate Boolean examples $x \in \{0,1\}^n$ from parameterized distributions. These worlds will contain:

- **base rules** that are true with high probability, such as $A \wedge B \rightarrow C$
- **exceptions** that break the rule on a controlled fraction of examples;
- **context variables** that alter rule validity, such as $A \rightarrow C$ but $A \wedge D \not\rightarrow C$.

Rule Language

We will focus on short, interpretable Boolean rules, such as:

- conjunction-to-literal rules: $(X_1 \wedge X_2) \rightarrow Y$;
- small disjunctive antecedents or alternative explanations;
- conditioned rules such as $(X_1 \wedge X_2) \rightarrow Y$ vs. $(X_1 \wedge X_2 \wedge Z) \rightarrow Y$.

Empirical Validity

For a candidate rule $\alpha \rightarrow \beta$, we will estimate empirical validity by

$$\widehat{\text{val}}(\alpha \rightarrow \beta) = \Pr[\beta = 1 \mid \alpha = 1]$$

computed from the sample whenever α occurs sufficiently often.

Non-Monotonic Reasoning

To study non-monotonicity, we will compare a rule's unconditional and conditional validity. For example, if a rule $\alpha \rightarrow \beta$ appears strong overall, we will evaluate whether

$$\Pr[\beta = 1 \mid \alpha = 1, \gamma = 1]$$

is larger, smaller, or qualitatively different for various contextual predicates γ .

The main question is whether adding context reveals exceptions or subgroup structure that is hidden at the aggregate level.

Abductive Explanation Search

We will also implement a simple abductive task. Given an observed outcome variable Y , we will search over a restricted class of candidate explanations H and score them using:

1. **plausibility:** how often H occurs;
2. **explanatory power:** how often Y occurs when H holds.

We will search over short conjunctions or disjunctions and identify explanations that are both frequent and strong enough to account for the observation.

Real Dataset 1: UCI Mushroom

The UCI Mushroom dataset is a good fit because it is categorical and explicitly does not admit a simple perfect edibility rule. We will convert its attributes into Boolean predicates and study rules such as whether combinations of odor, gill, and surface features are usually associated with poisonousness, and when these rules give structured exceptions.

This dataset will serve as a real-world test of approximate validity and exception-tolerant reasoning in a domain where simplistic rules are known to fail.

Real Dataset 2: EPA AQI

We will use the EPA AirData AQI files as a second case study. From daily or annual AQI summaries, we will construct Boolean predicates such as:

- `AQI_high`,
- `PM25_driven`,
- `summer`,
- `county_urban`,
- `coastal_region` or similar geographic subgroup labels if feasible.

The AQI case study will be used mainly to test contextual validity and subgroup-sensitive rules. For example, we will examine whether high-AQI events are usually associated with particular dominant pollutant categories, and whether such rules change across seasons, years, or broad geographic subgroups. This will provide a non-biological domain in which to examine approximate and non-monotonic reasoning.

Evaluation Metrics

We will evaluate the project mainly using the following three quantitative metrics.

1. **Empirical validity.** For a candidate rule $\alpha \rightarrow \beta$, we estimate its empirical validity by

$$\widehat{\text{val}}(\alpha \rightarrow \beta) = \frac{\#\{x : \alpha(x) = 1 \text{ and } \beta(x) = 1\}}{\#\{x : \alpha(x) = 1\}},$$

whenever $\#\{x : \alpha(x) = 1\} > 0$. This is the empirical conditional probability that the consequent holds given that the antecedent holds.

2. **Coverage.** We measure how often a rule is applicable by

$$\widehat{\text{cov}}(\alpha) = \frac{\#\{x : \alpha(x) = 1\}}{N},$$

where N is the total number of examples. This helps distinguish highly accurate rules that apply rarely from broadly useful rules.

3. **Context sensitivity.** To measure how much a rule changes after adding contextual information γ , we compare its base validity and refined validity:

$$\Delta_\gamma(\alpha \rightarrow \beta) = \widehat{\text{val}}((\alpha \wedge \gamma) \rightarrow \beta) - \widehat{\text{val}}(\alpha \rightarrow \beta).$$

Large positive or negative values of Δ_γ indicate that the rule is sensitive to added information, capturing the non-monotonic behavior we want to study.

We will also report several secondary indicators, including **abductive quality** (how plausible and predictive a candidate explanation is) and the **classical vs. approximate gap** (cases where exact validity fails but empirical validity remains high).

Deliverables

The intended deliverables are:

- a synthetic Boolean benchmark generator and experimental pipeline for evaluating approximate validity, contextual validity, and abductive explanations;
- experimental results on synthetic data and selected real datasets;
- a final five-page writeup describing the motivation, methods, experiments, and findings;
- a ten-minute final presentation summarizing the main ideas and results.

4 Project Timeline

- **Week 1:** finalize rule language, collect datasets, and implement Boolean preprocessing for Mushroom and AQI.
- **Week 2:** build synthetic Boolean generator with approximate rules and exceptions.
- **Week 3:** implement empirical validity, conditional validity, and explanation-search procedures.
- **Week 4:** run experiments on synthetic data; generate figures showing classical-vs.-approximate gaps and context effects.
- **Week 5:** run Mushroom and AQI case studies; refine analysis and prepare final report/presentation.

5 Backup Plan

If the full project proves too broad, we will reduce scope in the following order:

1. keep only one real dataset, most likely UCI Mushroom;
2. simplify abduction to a small search over conjunctions only;

3. focus the final analysis on empirical validity and non-monotonic conditional changes.

We want to preserve the main contribution, which is an empirical demonstration that approximate reasoning from examples can remain useful when classical deduction is too brittle.

6 Expected Outcomes

We expect to find that:

- many useful rules are not classically valid but still have high empirical validity on data;
- contextual refinement often produces non-monotonic effects, showing exceptions and subgroup-specific behavior;
- short abductive explanations can often be identified directly from examples;
- synthetic data will clearly show the gap between exact validity and approximate validity, while Mushroom and AQI will demonstrate that the same reasoning patterns can appear in real datasets.

More broadly, we expect the project to illustrate that reasoning from data is not merely a weaker substitute for logic, but can sometimes be the more appropriate framework when rules are approximate, exception-prone, and context-sensitive.

References

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